TEXCYCLE AI

AI-Powered Digital Marketplace for Upcycling Sri Lanka’s Textile Waste

PROJECT ID: R26-IT-081 | PANEL: IT-P06

- IT22281814 • IT22215338 • IT22221278 • IT22281678

Supervisor: Prof. Dasuni Nawinna Co-supervisor: Mr. Nelum Amarasena

01 RESEARCH TEAM

Four members with distinct, connected responsibilities



Supervisor: Prof. Dasuni Nawinna



Co-supervisor: Mr. Nelum Amarasena

IT22281814 | Sandeepa S.A.C.

Fabric Intelligence



IT22215338 | Siriwardhana N.S.T.N

Sustainability Intelligence



IT22221278 | Madushanka H.M.T.G

Traceability + Commercial Intelligence



IT22281678 | Thalagala T.A.D.V

Marketplace Predictive Analytics



PP2 PANEL COMMENTS & ACTIONS TAKEN

Project R26-IT-081 | Panel IT-P06 | Actions across all four individual components

MEMBER / COMPONENT	PP2 PANEL COMMENT	ACTION TAKEN FOR FINAL PRESENTATION
Member 01 **Fabric Intelligence**	• Progress was not sufficient for PP2. The implementation was basic, and the student could not clearly explain the model or outputs such as confidence and condition indicators.	• Added the image-to-prediction workflow; clarified the reported fabric/defect models, confidence meaning, output interpretation, and which condition values are model-based versus rule/formula-based. Model metrics are labelled for verification.
Member 02 **Sustainability Intelligence**	• Progress was not sufficient for PP2. The calculation of output values and the coding implementation were not clearly explained.	• Added the actual pipeline: eight input features → preprocessing → six XGBoost regressors → formula-derived indicators → XGBoost grade classifier and label encoder. Distinguished predicted outputs from formula-derived values and explained assumptions/limitations.
Member 03 **Commercial Intelligence**	• The panel did not see a clear understanding of the coding and implementation. Progress was considered insufficient for PP2.	• Clarified the pricing and buyer-recommendation workflow, the reported XGBoost pricing approach, traceability events and the distinction between model predictions and marketplace workflow. Reported metrics are identified for verification.
Member 04 **Marketplace Predictive Analytics**	• The student appeared unfamiliar with the coding and could not explain the code properly. Progress was considered insufficient for PP2.	• Added model-task summaries for listing success, sales-time prediction and market trends; described the input feature groups and clarified that each metric must be tied to its target, test protocol and data quality.

02 PROJECT OVERVIEW

One platform, four research contributions

TexCycle AI connects textile assessment to sustainability information and marketplace decision support.

IDENTIFY

Understand the uploaded material and visible defects.

ASSESS

Estimate sustainability-related indicators with documented assumptions.

CONNECT + PREDICT

Support price decisions, buyer discovery and marketplace forecasting.

Core contribution

A role-based digital marketplace where AI outputs help users make more informed decisions about textile-waste reuse and upcycling.

03 INTRODUCTION

Why textile waste needs a smarter route to value

MATERIAL INFORMATION

Fabric type and condition may be unclear from a basic listing.

VALUE UNCERTAINTY

Sellers may need support estimating value and interpreting sustainability indicators.

FRAGMENTED DISCOVERY

Relevant buyers and market signals can be difficult to connect consistently.

Research alignment: SDG 12 — Responsible Consumption and Production; SDG 8 — Decent Work and Economic Growth.

04 RESEARCH PROBLEM & GAP

The challenge is a chain of connected decisions

PROBLEM A

Material category and condition are not always represented in a structured way.

PROBLEM B

Sustainability indicators and indicative values need clear calculation and interpretation.

PROBLEM C

Buyer matching and marketplace predictions need to be connected to listing information.

Research question

How can AI-assisted textile assessment and marketplace intelligence support informed reuse, upcycling and trading decisions?

05 OBJECTIVES

One main objective is supported by four component objectives

MAIN OBJECTIVE

Develop an AI-powered digital marketplace that helps users assess textile materials and make more informed sustainability and commercial decisions.

01 Identify material characteristics

Fabric classification and defect/condition assessment

02 Estimate sustainability indicators

Scores, grades and environmental estimates with explicit assumptions

03 Support commercial decisions

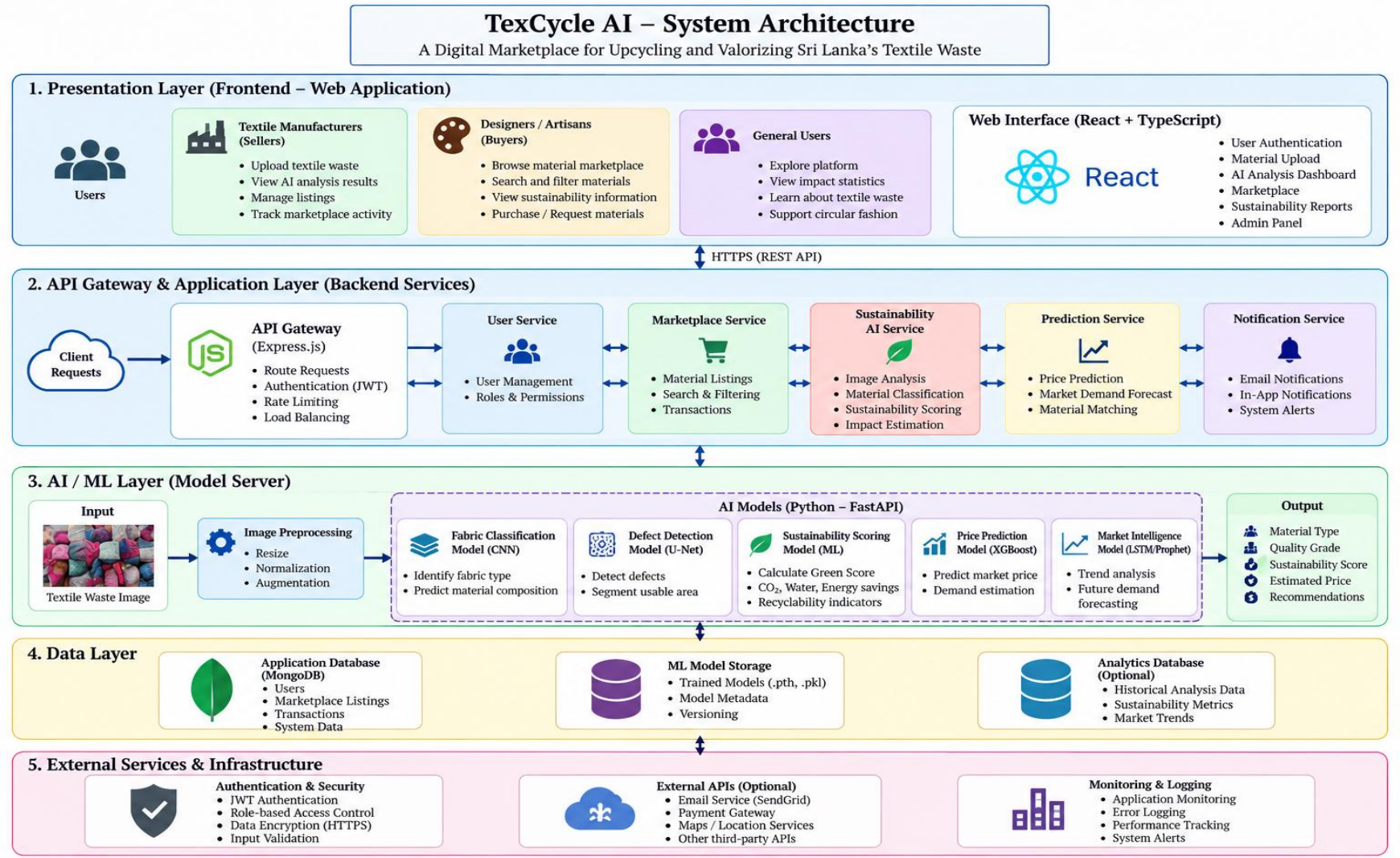
Indicative pricing, buyer recommendation and traceability

04 Predict marketplace patterns

Listing success, sales time, market trend, and demand were implemented

06 FULL SYSTEM ARCHITECTURE

How the website, backend, AI services, and data work together



07 USER ROLES & MARKETPLACE WORKFLOW

The same platform serves different users

SELLER

- Upload material
- Analyses and list
- View predictions
- Manage sales

BUYER

- Browse materials
- Choose quantity
- Place orders
- Track and review

ADMIN

- Manage platform activity
- Review users/listings
- Monitor orders and reports

SUPER ADMIN

- Platform-level access
- Administrative governance

Shared services: authentication • marketplace listings • orders • notifications • reviews/ratings • history (as implemented)

08 RESEARCH METHODOLOGY

How the team moved from research to a working prototype

1	**Literature + requirements**	Identify problem, research gap and component targets
2	**Dataset preparation**	Review industry-visit notes and global/Sri Lankan reports where available; prepare and document component-specific datasets, labels and features. Identify any synthetic or derived records separately.
3	**Model development**	Train task-specific classifiers, regressors or segmentation models
4	**Evaluation**	Use task-appropriate metrics on held-out data
5	**Integration**	Connect inference/results to API and role-based website workflows
6	**Testing + analysis**	Test workflows, document limitations and interpret results

09 INDIVIDUAL RESEARCH CONTRIBUTIONS

Four components, four different research questions

MEMBER 01 | Fabric Intelligence

What material is it? What defects can be identified?

MEMBER 02 | Sustainability Intelligence

What environmental/sustainability indicators can be estimated?

MEMBER 03 | Commercial Intelligence

What price and buyer options may help the seller?

MEMBER 04 | Marketplace Predictive Analytics

What marketplace outcomes or trends can be predicted?

10 MEMBER 01 | IT22281814 | Sandeepa S.A.C.



Fabric Intelligence — research focus & solution

RESEARCH INPUT

Uploaded textile image; image validation and preprocessing.

MODELS REPORTED

EfficientNet-B3 for fabric classification; MobileNetV2 for defect-related classification; U-Net for segmentation only if confirmed in final implementation.

USER OUTPUT

Fabric category + confidence; defect result / mask where integrated; condition indicators only where implemented.

Reported result from earlier project notes: EfficientNet-B3 test accuracy 89.68%; MobileNetV2 result 90.87% (verify exact metric and task).

11 MEMBER 01 | METHOD



From textile image to a useful result

Upload

Image

Preprocess

Resize / normalize

Inference

Trained model

Post-process

Class/mask

Display

Prediction

12 MEMBER 02 | IT22215338 | Siriwardhana N.S.T.N



Sustainability Intelligence — model outputs and calculations

01 Circularity score
XGBoost Regressor
Model/target scale

02 Sustainability grade
XGBoost Classifier + label encoder
Grade model or mapping

03 Recyclability score
XGBoost Regressor
Exact implemented formula

04 CO_2 saved (estimate)
XGBoost Regressor
Units + baseline factor

05 Water saved (estimate)
XGBoost Regressor
Units + baseline factor

06 Carbon reduction %
XGBoost Regressor
Boundary + units

07 Water reduction %
XGBoost Regressor
Inputs combined

08 Derived indicators
Energy, landfill, SDG & economic proxy scores
Reuse/recycle/upcycle if implemented

Earlier notes report R^2 = 0.9389–0.9887 across sustainability experiments. Verify each value against its exact model, target and held-out test set before presenting it as a final result.

13 MEMBER 02 | METHOD



Member 02 method: inputs → predictions → derived indicators

STAGE 1 — MODEL-BASED

Inputs: fabric type, weight (kg), health score, repairability, condition, district, province and industry type. A saved preprocessor transforms inputs; six XGBoost regressors predict six indicators. Dataset preparation should be explained with the actual industry-visit/report sources used for this component.

STAGE 2 — FORMULA-BASED

Code-derived examples: energy_saved_kwh = CO_2_saved_kg × 1.5; landfill_diverted_kg = weight_kg × 0.95; SDG proxy = mean(circularity, recyclability). These are estimates/proxies, not measured impacts.

STAGE 3 — GRADE CLASSIFICATION

A separate XGBoost classifier predicts the sustainability grade from the prepared feature row; a saved label encoder converts the class to its grade label.

- State input features and units.
- Explain each predicted output and derived formula.
- Disclose baselines and assumptions.
- Outputs are clipped to defined ranges and rounded to 2 decimals.
- No public hosting or independent environmental certification is claimed.

14 MEMBER 03 | IT22221278 | Madushanka H.M.T.G



Commercial Intelligence — pricing and buyer recommendation

PRICE MODEL

XGBoost Regressor
MAE 12.24 • RMSE 15.40
R^2 0.9590 (prior reported result)

BUYER RECOMMENDATION

Prior notes describe Random Forest + XGBoost. Reported matching result: 83.56%; verify how matching quality was measured.

TRACEABILITY

Track analysis/listing/order events and history where implemented. Traceability is a system workflow, not automatically an ML model.

Commercial flow

Material + condition + quantity + market features → feature preparation → price estimate/buyer suggestions → seller decision.

15 MEMBER 03 | METHOD



Traceability links the material journey

Material

Upload / details

Analysis

Outputs / history

Listing

Published offer

Order

Status / delivery

Feedback

Review / rating if implemented

Why it matters

A more structured history can support visibility and accountability.

What it cannot prove alone

Records alone do not verify origin or environmental impact.



16 MEMBER 04 | IT22281678 | Thalagala T.A.D.V

Marketplace Predictive Analytics — model tasks

LISTING SUCCESS

XGBoost Classifier
Reported accuracy: 83.65%
Define the success label.

SALES TIME

XGBoost Regressor
Prior notes: MAE 14.53 days; RMSE 20.87; R^2 0.76

MARKET TREND

Increasing / Stable / Decreasing classes. Prior reported accuracy: 98.72%; verify the test protocol.

DEMAND

Present separately only if a distinct trained model and evaluation exist.

17 MEMBER 04



Marketplace dataset and feature groups

Material/condition

fabric_type • condition • health_score • defect_area • repairability • remaining_lifespan • weight_kg

Sustainability/price

sustainability_score • green_score • co2_saving • water_saving • predicted_price • average_market_price

Market context

buyer_type • demand_level • listing_month • season • district • province • competition • economic_index

Seller/engagement

seller_rating • weekly_views • wishlist_count • click_rate • seller_experience • listing_status • days_to_sell

18 INTEGRATED USER JOURNEY

How the four components work together on the website

01 • Upload	Seller uploads an image and enters material details.
02 • Identify	Member 01 model classifies fabric and analyses defects where supported.
03 • Assess	Member 02 produces sustainability indicators and environmental estimates.
04 • Commercialise	Member 03 provides indicative price, buyer recommendations and traceability.
05 • Predict + trade	Member 04 provides marketplace predictions; users continue through listings and orders.

Claim automatic data hand-offs only where the current code implements them; otherwise, describe the components as connected through shared workflows.

19 SUMMARY

Report the metric that matches the model task

Task	Model / method	Prior reported result	Required verification
Fabric classification	EfficientNet-B3	Accuracy 89.68%	Confirm classes / test split
Defect model	MobileNetV2 / U-Net	90.87% reported	Confirm exact metric and model
Price prediction	XGBoost regressor	MAE 12.24 • RMSE 15.40 • R^2 0.9590	Confirm currency/unit
Listing success	XGBoost classifier	Accuracy 83.65%	Confirm success label
Sales time	XGBoost regressor	MAE 14.53 days • RMSE 20.87 • R^2 0.76	Weak explanatory power
Market trend	Classifier	Accuracy 98.72% reported	Check generated labels / leakage
Buyer recommendation	RF + XGBoost approach	83.56% reported	Define evaluation protocol
Sustainability models	Regressors / grade model	R^2 range 0.9389–0.9887	Map each value to target

21 LIMITATIONS & NEXT VALIDATION

Limitations, testing status and future improvements

DATA SOURCES

Dataset sources differ by component. Industry visits and global/Sri Lankan industry reports informed dataset preparation where documented. The marketplace predictive dataset includes synthetic/prototype records; therefore, real buyer behaviour, transaction prices and seasonal demand still require validation.

MODEL VALIDATION

Reported metrics must be verified for each model and target. Check for data leakage and evaluate on independent held-out data. Report only metrics supported by the relevant test evidence.

DEPLOYMENT & IMPACT

The prototype was tested locally and reviewed by external users/stakeholders. Feedback was positive, and users expressed satisfaction with the system. The prototype has not been publicly hosted. Independent environmental-impact certification remains future work. Next steps: secure deployment and broader validation.

RESULTS, TESTING & FUTURE IMPROVEMENTS

Prototype status — tested locally and reviewed by external users/stakeholders. Feedback was positive and users expressed satisfaction. The prototype has not been publicly hosted.

Earlier notes report sustainability R^2 values from 0.9389–0.9887. Verify each model, target, dataset source, split and held-out result before treating these as final.

Functional testing

Demonstrate locally: input validation → preprocessing → model predictions → derived indicators → grade response. Add pass/fail evidence only from tests actually run.

External users/stakeholders reviewed and tested the locally run prototype. Feedback was positive, and users expressed satisfaction with the system. Keep available test notes, feedback and screenshots as supporting evidence. No numeric satisfaction rate or independent benchmark score is claimed.

Limitations

Not publicly hosted; environmental savings are estimates based on model outputs and code-defined factors; prototype/synthetic data may not represent real Sri Lankan market behaviour.

Future improvements

Deploy securely; broaden testing with more sellers, buyers and textile stakeholders; collect real listings and transactions; validate environmental factors; report MAE/RMSE/R^2 or accuracy/precision/recall/F1 as appropriate.

REFERENCES

Reference slide — align with the approved thesis bibliography

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Thank You!